MATA37H3 — Week 05

Improper Integrals: Type 1, Type 2, p-Integrals and the Comparison Tests

Recap and orientation

Last week completed the integration-techniques toolkit (trigonometric integrals, partial fractions, the Weierstrass substitution). This week those techniques are used inside **limits**: an **improper integral** extends $\int_a^b f(x)\,dx$ to an infinite interval (Type 1) or to an integrand that blows up at an endpoint or interior point (Type 2). Each is defined as a limit of proper integrals, and it **converges** only if that limit is finite.

Why this matters

When an antiderivative is hard or impossible to find, the **Basic Comparison Test** and **Limit Comparison Test** decide convergence by comparing with a benchmark p-integral. Series convergence tests in Weeks 6-14 are the discrete versions of exactly these arguments.

DEFINITION **Type-1 Improper Integral (Infinite Interval)**

If f is continuous on $[a,+\infty)$, then

$$\int_a^{+\infty} f(x)\,dx = \lim_{t\to+\infty}\int_a^t f(x)\,dx,$$

provided the limit exists. Similarly $\displaystyle\int_{-\infty}^b f(x)\,dx = \lim_{t\to-\infty}\int_t^b f(x)\,dx$. For $(-\infty,+\infty)$, split at any real a:

$$\int_{-\infty}^{+\infty} f(x)\,dx = \int_{-\infty}^a f(x)\,dx + \int_a^{+\infty} f(x)\,dx,$$

which converges only if **both** pieces converge. If the limit exists and is finite the integral **converges**; otherwise it **diverges**.

DEFINITION **Type-2 Improper Integral (Discontinuous Integrand)**

If f is continuous on $[a,b)$ with a discontinuity at b, then $\displaystyle\int_a^b f(x)\,dx = \lim_{t\to b^-}\int_a^t f(x)\,dx$. If the discontinuity is at a, use $\displaystyle\lim_{t\to a^+}\int_t^b f(x)\,dx$. If f is discontinuous at an interior point c, split:

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx,$$

which converges only if **both** pieces converge.

KEY FACT **The p-Integrals**

$$\int_1^{+\infty}\tfrac{1}{x^p}\,dx \text{ converges } \iff p>1, \qquad \int_0^1\tfrac{1}{x^p}\,dx \text{ converges } \iff p<1.$$

Consequently $\displaystyle\int_0^{+\infty} \frac{1}{x^p}\,dx$ diverges for every p, since it splits into the two integrals above and no single p makes both converge.

KEY FACT **Basic Comparison Test (BCT)**

Let f, g be continuous on $[a, +\infty)$ with $0 \le f(x) \le g(x)$ for all $x \ge a$.

- If $\displaystyle\int_a^{+\infty} f(x)\,dx$ diverges, then $\displaystyle\int_a^{+\infty} g(x)\,dx$ diverges ("small diverges $\Rightarrow$ big diverges").
- If $\displaystyle\int_a^{+\infty} g(x)\,dx$ converges, then $\displaystyle\int_a^{+\infty} f(x)\,dx$ converges ("big converges $\Rightarrow$ small converges").

The other two implications are false in general. The inequality only needs to hold eventually.

KEY FACT **Limit Comparison Test (LCT)**

Let f, g be positive and continuous on $[a, +\infty)$ with $L = \displaystyle\lim_{x\to+\infty} \frac{f(x)}{g(x)}$.

- If $0 < L < +\infty$, then $\displaystyle\int_a^{+\infty} f(x)\,dx$ and $\displaystyle\int_a^{+\infty} g(x)\,dx$ both converge or both diverge.
- The same statement holds for Type-2 integrals, taking the limit as x approaches the singular endpoint (for example $x \to 0^+$).

Worked Examples

EXAMPLE 1

Level: Foundational. Type 1 integral with partial fractions

Evaluate $\displaystyle\int_1^{+\infty} \frac{1}{x^2+x}\,dx$.

Step 1. Classify the integral. The upper limit is $+\infty$, so this is a Type 1 improper integral. By definition we replace the infinite limit by a finite number t and take a limit:

$$\int_1^{+\infty} \frac{1}{x^2+x}\,dx = \lim_{t\to+\infty} \int_1^t \frac{1}{x^2+x}\,dx.$$

The integrand is continuous on $[1, t]$ because the denominator $x^2 + x$ is positive there.

Step 2. To find an antiderivative, factor the denominator: $x^2 + x = x(x+1)$. The integrand is a rational function whose denominator is a product of distinct linear factors, so we use partial fractions:

$$\frac{1}{x(x+1)} = \frac{A}{x} + \frac{B}{x+1}.$$

Step 3. Multiply both sides by the common denominator $x(x+1)$ to clear the fractions:

$1 = A(x+1) + Bx$. This equation must hold for every x.

Step 4. Substitute $x = 0$, which makes the B term vanish: $1 = A(0+1) + B \cdot 0 = A$, so $A = 1$.

Step 5. Substitute $x = -1$, which makes the A term vanish: $1 = A \cdot 0 + B(-1) = -B$, so $B = -1$.

Step 6. Check the decomposition by recombining: $\dfrac{1}{x} - \dfrac{1}{x+1} = \dfrac{(x+1) - x}{x(x+1)} = \dfrac{1}{x(x+1)}$. This equals the original integrand, so $\dfrac{1}{x^2+x} = \dfrac{1}{x} - \dfrac{1}{x+1}$.

Step 7. Integrate term by term. $\int \dfrac{1}{x}\,dx = \ln|x|$. For $\int \dfrac{1}{x+1}\,dx$ let $u = x+1$, so $du = dx$ and the integral is $\int \dfrac{du}{u} = \ln|u| = \ln|x+1|$. Therefore an antiderivative is $\ln|x| - \ln|x+1|$.

Step 8. On the interval $[1, t]$ both x and $x+1$ are positive, so the absolute value bars can be dropped:

$$\int_1^t \left(\frac{1}{x} - \frac{1}{x+1}\right) dx = \Big[\ln x - \ln(x+1)\Big]_1^t.$$

Step 9. Evaluate at the upper limit $x = t$: $\ln t - \ln(t+1)$. Evaluate at the lower limit $x = 1$: $\ln 1 - \ln 2 = 0 - \ln 2 = -\ln 2$.

Step 10. Subtract (upper minus lower):

$$\int_1^t \frac{1}{x^2+x}\,dx = \big(\ln t - \ln(t+1)\big) - \big(-\ln 2\big) = \ln t - \ln(t+1) + \ln 2.$$

Step 11. Combine the first two logarithms with the quotient rule $\ln a - \ln b = \ln \dfrac{a}{b}$:

$$\ln t - \ln(t+1) = \ln \frac{t}{t+1}.$$

Step 12. Take the limit $t \to +\infty$. Divide numerator and denominator of $\dfrac{t}{t+1}$ by t: $\dfrac{t}{t+1} = \dfrac{1}{1+\frac{1}{t}} \to \dfrac{1}{1+0} = 1$. Since ln is continuous at 1, $\ln \dfrac{t}{t+1} \to \ln 1 = 0$.

Step 13. Therefore the limit is $0 + \ln 2 = \ln 2$, which is finite, so the integral converges.

Step 14. Numerical check with $t = 1000$: $\ln \dfrac{1000}{1001} \approx -0.0009995$, so the partial integral is about $-0.0009995 + 0.693147 \approx 0.692148$, already very close to $\ln 2 \approx 0.693147$.

Answer. Converges to $\ln 2$

EXAMPLE 2

Level: Intermediate. The p-integral derived from the definition

Show from the definition that $\displaystyle\int_1^{+\infty} \frac{1}{x^p}\,dx$ converges if and only if $p > 1$, and find its value when it converges.

Step 1. Write the integral as a limit:

$$\int_1^{+\infty} \frac{1}{x^p}\,dx = \lim_{t\to+\infty} \int_1^t x^{-p}\,dx.$$

The antiderivative of x^{-p} depends on whether $-p = -1$, so we treat $p \neq 1$ and $p = 1$

separately.

Step 2. Case $p \neq 1$. Use the power rule $\int x^k\,dx = \dfrac{x^{k+1}}{k+1}$ with $k = -p$ (allowed because $k \neq -1$):

$$\int x^{-p}\,dx = \frac{x^{1-p}}{1-p}.$$

Check: $\dfrac{d}{dx}\left[\dfrac{x^{1-p}}{1-p}\right] = \dfrac{(1-p)x^{-p}}{1-p} = x^{-p}$.

Step 3. Evaluate from 1 to t, using $1^{1-p} = 1$:

$$\int_1^t x^{-p}\,dx = \frac{t^{1-p}-1}{1-p}.$$

Step 4. Subcase $p > 1$. Then $1-p < 0$, so write $1-p = -(p-1)$ with $p-1 > 0$. Hence $t^{1-p} = \dfrac{1}{t^{p-1}}$, and since $t^{p-1} \to +\infty$ as $t \to +\infty$, we get $t^{1-p} \to 0$.

Step 5. Then the limit is $\dfrac{0-1}{1-p} = \dfrac{-1}{1-p}$. Since $1-p = -(p-1)$, this equals $\dfrac{1}{p-1}$, a finite number. So for $p > 1$ the integral converges to $\dfrac{1}{p-1}$.

Step 6. Spot-check with $p = 2$: $\int_1^t x^{-2}\,dx = \left[-\dfrac{1}{x}\right]_1^t = 1 - \dfrac{1}{t} \to 1$, and the formula gives $\dfrac{1}{2-1} = 1$.

Step 7. Subcase $p < 1$. Then $1-p > 0$, so t^{1-p} is a positive power of t and $t^{1-p} \to +\infty$. The numerator $t^{1-p} - 1 \to +\infty$, and the denominator $1-p$ is a fixed positive number, so the quotient tends to $+\infty$. The integral diverges.

Step 8. Spot-check with $p = \dfrac{1}{2}$: $\int_1^t x^{-1/2}\,dx = \left[2\sqrt{x}\right]_1^t = 2\sqrt{t} - 2 \to +\infty$.

Step 9. Case $p = 1$. Here $\int \dfrac{1}{x}\,dx = \ln|x|$, so

$$\int_1^t \frac{1}{x}\,dx = \ln t - \ln 1 = \ln t.$$

Step 10. The function $\ln t$ is increasing and unbounded: for any bound M, $\ln t > M$ once $t > e^M$. So $\ln t \to +\infty$ and the integral diverges.

Step 11. Combine the cases: for $p > 1$ the limit is the finite number $\dfrac{1}{p-1}$; for $p < 1$ and for $p = 1$ the limit is $+\infty$. Therefore the integral converges exactly when $p > 1$.

Step 12. Final check with $p = 3$: the formula gives $\dfrac{1}{3-1} = \dfrac{1}{2}$, and directly $\left[-\dfrac{1}{2x^2}\right]_1^{+\infty} = 0 - \left(-\dfrac{1}{2}\right) = \dfrac{1}{2}$.

Answer. Converges if and only if $p > 1$, with value $\dfrac{1}{p-1}$; diverges for $p \leq 1$

EXAMPLE 3

Level: Intermediate. Both limits infinite: splitting at a point

Determine whether $\displaystyle\int_{-\infty}^{+\infty} x\,e^{-x^2}\,dx$ converges, and find its value if it does.

Step 1. Both limits of integration are infinite, so the definition requires a split at any convenient real number a:

$$\textstyle\int_{-\infty}^{+\infty} f(x)\,dx = \int_{-\infty}^{a} f(x)\,dx + \int_{a}^{+\infty} f(x)\,dx,$$

and the whole integral converges only if each of the two pieces converges on its own. Choose $a = 0$ because it simplifies the exponential.

Step 2. First find an antiderivative of xe^{-x^2}. The exponent $-x^2$ has derivative $-2x$, which is a constant multiple of the extra factor x, so use the substitution $u = -x^2$.

Step 3. Differentiate: $du = -2x\,dx$, so $x\,dx = -\dfrac{du}{2}$.

Step 4. Substitute: $\displaystyle\int xe^{-x^2}\,dx = -\frac{1}{2}\int e^u\,du = -\frac{1}{2}e^u = -\frac{1}{2}e^{-x^2}$.

Step 5. Check by differentiating with the chain rule: $\dfrac{d}{dx}\left[-\dfrac{1}{2}e^{-x^2}\right] = -\dfrac{1}{2}e^{-x^2}\cdot(-2x) = xe^{-x^2}$.

Step 6. Right piece. By definition $\displaystyle\int_0^{+\infty} xe^{-x^2}\,dx = \lim_{t\to+\infty}\int_0^t xe^{-x^2}\,dx$, and

$$\textstyle\int_0^t xe^{-x^2}\,dx = \Big[-\dfrac{1}{2}e^{-x^2}\Big]_0^t = -\dfrac{1}{2}e^{-t^2} - \left(-\dfrac{1}{2}e^0\right) = \dfrac{1}{2} - \dfrac{1}{2}e^{-t^2}.$$

Step 7. As $t\to+\infty$, $t^2\to+\infty$, so $e^{-t^2}\to 0$. The right piece converges to $\dfrac{1}{2}$.

Step 8. Left piece. By definition $\displaystyle\int_{-\infty}^0 xe^{-x^2}\,dx = \lim_{s\to-\infty}\int_s^0 xe^{-x^2}\,dx$, and

$$\textstyle\int_s^0 xe^{-x^2}\,dx = \Big[-\dfrac{1}{2}e^{-x^2}\Big]_s^0 = -\dfrac{1}{2} - \left(-\dfrac{1}{2}e^{-s^2}\right) = -\dfrac{1}{2} + \dfrac{1}{2}e^{-s^2}.$$

Step 9. As $s\to-\infty$, the square $s^2\to+\infty$ (a large negative number squared is a large positive number), so $e^{-s^2}\to 0$. The left piece converges to $-\dfrac{1}{2}$.

Step 10. Both pieces converge to finite numbers, so the full integral converges, and its value is the sum:

$$-\frac{1}{2} + \frac{1}{2} = 0.$$

Step 11. Consistency check with a different split point $a = 1$: $\displaystyle\int_1^{+\infty} = \lim_{t\to\infty}\left[-\frac{1}{2}e^{-x^2}\right]_1^t =$ $\dfrac{1}{2}e^{-1}$ and $\displaystyle\int_{-\infty}^1 = \left[-\frac{1}{2}e^{-x^2}\right]_{-\infty}^1 = -\frac{1}{2}e^{-1} - 0$. The sum is again 0.

Step 12. Check of the signs: the integrand $f(x) = xe^{-x^2}$ satisfies $f(-x) = -xe^{-x^2} = -f(x)$ (it is odd), so the left piece is the negative of the right piece, matching $-\dfrac{1}{2}$ and $+\dfrac{1}{2}$.

Answer. Converges, and $\displaystyle\int_{-\infty}^{+\infty} xe^{-x^2}\,dx = 0$

EXAMPLE 4

Level: Foundational. Type 2 integral with an endpoint singularity

Evaluate $\displaystyle\int_0^4 \frac{1}{\sqrt{4-x}}\,dx$.

Step 1. Locate any discontinuity. The integrand $\dfrac{1}{\sqrt{4-x}}$ is defined and continuous when $4-x>0$, that is for $x<4$. At $x=4$ the denominator is 0 and the function grows without bound (a vertical asymptote). So the discontinuity is at the upper endpoint and the integral is Type 2.

Step 2. By definition, stop short of the bad endpoint and take a limit:

$$\int_0^4 \frac{1}{\sqrt{4-x}}\,dx = \lim_{t\to 4^-}\int_0^t \frac{1}{\sqrt{4-x}}\,dx.$$

The notation $t\to 4^-$ means t approaches 4 from the left, which keeps t inside $[0,4)$.

Step 3. Rewrite the integrand with a negative exponent: $\dfrac{1}{\sqrt{4-x}} = (4-x)^{-1/2}$.

Step 4. Use the substitution $u=4-x$. Differentiating, $du=-dx$, so $dx=-du$.

Step 5. Substitute: $\int (4-x)^{-1/2}\,dx = -\int u^{-1/2}\,du$.

Step 6. Apply the power rule $\int u^k\,du = \dfrac{u^{k+1}}{k+1}$ with $k=-\dfrac{1}{2}$, so $k+1=\dfrac{1}{2}$:

$$\int u^{-1/2}\,du = \frac{u^{1/2}}{\frac{1}{2}} = 2\sqrt{u}.$$

Step 7. Include the minus sign from step 5 and return to x: $\displaystyle\int \frac{1}{\sqrt{4-x}}\,dx = -2\sqrt{u} = -2\sqrt{4-x}$.

Step 8. Check by differentiating: $\dfrac{d}{dx}\left[-2(4-x)^{1/2}\right] = -2\cdot\dfrac{1}{2}(4-x)^{-1/2}\cdot(-1) = (4-x)^{-1/2}$, which is the integrand.

Step 9. Evaluate on $[0,t]$: at the upper limit, $-2\sqrt{4-t}$; at the lower limit, $-2\sqrt{4-0} = -2\cdot 2 = -4$.

Step 10. Subtract:

$$\int_0^t \frac{1}{\sqrt{4-x}}\,dx = -2\sqrt{4-t} - (-4) = 4 - 2\sqrt{4-t}.$$

Step 11. Take the limit $t\to 4^-$. Then $4-t\to 0^+$, so $\sqrt{4-t}\to 0$, and the expression tends to $4-0=4$.

Step 12. The limit is finite, so the integral converges to 4, even though the integrand is unbounded near $x=4$.

Step 13. Numerical check: for $t=3.9999$, $4-t=0.0001$ and $\sqrt{0.0001}=0.01$, so the partial integral is $4-0.02=3.98$, close to 4.

Answer. Converges to 4

EXAMPLE 5

Level: Intermediate. Interior discontinuity: split at the bad point

Evaluate

$$\int_{-1}^{1} \frac{1}{x^3}\,dx,$$

or show that it diverges.

Step 1. Look for points where the integrand $f(x) = \dfrac{1}{x^3}$ fails to be continuous on $[-1, 1]$. The denominator x^3 is zero only at $x = 0$, and $\left|\dfrac{1}{x^3}\right| \to \infty$ as $x \to 0$. So f has a vertical asymptote at $x = 0$.

Step 2. The point $x = 0$ lies strictly inside the interval $(-1, 1)$, not at an endpoint. So this is a Type-2 improper integral with an interior discontinuity.

Step 3. The Fundamental Theorem of Calculus requires the integrand to be continuous on the whole closed interval, so it cannot be applied across $x = 0$. Instead we use the definition: split the integral at the discontinuity $c = 0$,

$$\int_{-1}^{1} \frac{1}{x^3}\,dx = \int_{-1}^{0} \frac{1}{x^3}\,dx + \int_{0}^{1} \frac{1}{x^3}\,dx,$$

and treat each piece as its own limit. The original integral converges only if both pieces converge; if either one diverges, the whole integral diverges.

Step 4. Take the first piece. On $[-1, 0)$ the function is continuous and the only trouble is at the right endpoint 0, so by the Type-2 definition

$$\int_{-1}^{0} \frac{1}{x^3}\,dx = \lim_{t \to 0^-} \int_{-1}^{t} x^{-3}\,dx.$$

Step 5. Find an antiderivative with the power rule $\int x^n\,dx = \dfrac{x^{n+1}}{n+1}$ for $n = -3$. Here $n + 1 = -2$, so $\int x^{-3}\,dx = \dfrac{x^{-2}}{-2} = -\dfrac{1}{2x^2}$.

Step 6. Check the antiderivative by differentiating: $\dfrac{d}{dx}\left[-\dfrac{1}{2}x^{-2}\right] = -\dfrac{1}{2} \cdot (-2)x^{-3} = x^{-3}$, which is the integrand.

Step 7. Evaluate between the limits -1 and t (note $t < 0$, but $t^2 > 0$):

$$\int_{-1}^{t} x^{-3}\,dx = \left[-\frac{1}{2x^2}\right]_{-1}^{t} = -\frac{1}{2t^2} - \left(-\frac{1}{2(-1)^2}\right) = -\frac{1}{2t^2} + \frac{1}{2}.$$

Step 8. Take the limit $t \to 0^-$. Then $t^2 \to 0^+$, so $\dfrac{1}{2t^2} \to +\infty$ and therefore $-\dfrac{1}{2t^2} + \dfrac{1}{2} \to -\infty$.

Step 9. The first piece diverges (to $-\infty$). By the definition in step 3, a single divergent piece makes the whole improper integral diverge.

Step 10. For completeness, the second piece also diverges: $\lim\limits_{t \to 0^+} \displaystyle\int_{t}^{1} x^{-3}\,dx = \lim\limits_{t \to 0^+} \left[-\dfrac{1}{2x^2}\right]_{t}^{1} = \lim\limits_{t \to 0^+} \left(-\dfrac{1}{2} + \dfrac{1}{2t^2}\right) = +\infty$. The two infinite contributions $-\infty$ and $+\infty$ do not cancel; each piece must converge on its own.

Step 11. Cross-check with the p-integral rule $\int_0^1 \dfrac{1}{x^p}\,dx$ converges if and only if $p < 1$. Here $p = 3$, which is not less than 1, so the piece on $[0, 1]$ diverges, in agreement with the computation above (and the same holds for the mirror piece on $[-1, 0]$).

Step 12. Conclusion: $\displaystyle\int_{-1}^{1} \frac{1}{x^3}\,dx$ diverges.

Answer. The integral diverges: $\displaystyle\int_{-1}^{0} \frac{1}{x^3}\,dx = -\infty$ and $\displaystyle\int_{0}^{1} \frac{1}{x^3}\,dx = +\infty$.

EXAMPLE 6

Level: Intermediate. Basic Comparison Test: small diverges, so big diverges

Determine whether

$$\int_1^{+\infty} \frac{1+\cos^2 x}{x^{\frac{2}{3}}}\,dx$$

converges or diverges.

Step 1. The antiderivative of $\dfrac{1+\cos^2 x}{x^{2/3}}$ is not available in closed form, so we cannot use the definition directly. We compare the integrand with a simpler function whose integral we can compute. Write $f(x) = \dfrac{1+\cos^2 x}{x^{\frac{2}{3}}}$.

Step 2. Check that f is continuous and nonnegative on $[1, +\infty)$. The numerator $1+\cos^2 x$ is continuous, and the denominator $x^{\frac{2}{3}}$ is continuous and at least 1 for $x \geq 1$, so it never vanishes. Also $1+\cos^2 x > 0$, so $f(x) > 0$.

Step 3. Bound the numerator from below. A square is never negative, so $\cos^2 x \geq 0$ for every real x. Adding 1 to both sides gives $1+\cos^2 x \geq 1$.

Step 4. Divide both sides of $1+\cos^2 x \geq 1$ by $x^{\frac{2}{3}}$. Because $x^{\frac{2}{3}} > 0$ for $x \geq 1$, dividing by it keeps the inequality direction:

$$f(x) = \frac{1+\cos^2 x}{x^{\frac{2}{3}}} \geq \frac{1}{x^{\frac{2}{3}}} = g(x) > 0 \qquad \text{for all } x \geq 1.$$

Step 5. So $0 \leq g(x) \leq f(x)$ on $[1, +\infty)$, with $g(x) = x^{-\frac{2}{3}}$ continuous. This is the setting of the Basic Comparison Test, with g playing the role of the smaller function.

Step 6. Decide the behaviour of $\displaystyle\int_1^{+\infty} g(x)\,dx$. This is a p-integral with $p = \frac{2}{3}$, and $\frac{2}{3} \leq 1$, so it should diverge. Confirm from the definition with the power rule $\int x^n\,dx = \dfrac{x^{n+1}}{n+1}$, $n = -\dfrac{2}{3}$, $n+1 = \dfrac{1}{3}$:

$$\int_1^t x^{-\frac{2}{3}}\,dx = \left[\frac{x^{\frac{1}{3}}}{\frac{1}{3}}\right]_1^t = \left[3x^{\frac{1}{3}}\right]_1^t = 3t^{\frac{1}{3}} - 3.$$

Step 7. Check the antiderivative: $\dfrac{d}{dx}\left[3x^{\frac{1}{3}}\right] = 3 \cdot \dfrac{1}{3}x^{-\frac{2}{3}} = x^{-\frac{2}{3}}$.

Step 8. Take the limit as $t \to +\infty$: $t^{\frac{1}{3}} \to +\infty$, so $3t^{\frac{1}{3}} - 3 \to +\infty$. Hence $\displaystyle\int_1^{+\infty} \frac{1}{x^{\frac{2}{3}}}\,dx$ diverges.

Step 9. State the Basic Comparison Test, part (i): if $0 \le g(x) \le f(x)$ on $[a, +\infty)$ and $\displaystyle\int_a^{+\infty} g(x)\,dx = +\infty$, then $\displaystyle\int_a^{+\infty} f(x)\,dx = +\infty$ (the smaller function diverges, so the larger one diverges).

Step 10. Make the conclusion concrete without quoting the theorem: for every $t \ge 1$, $\displaystyle\int_1^t f(x)\,dx \ge \int_1^t g(x)\,dx = 3t^{\frac{1}{3}} - 3$, and the right side tends to $+\infty$. A quantity that is at least something tending to $+\infty$ also tends to $+\infty$.

Step 11. Spot check of the inequality at $x = 1$: $f(1) = \dfrac{1 + \cos^2 1}{1} \approx 1.292$ and $g(1) = 1$, so $f(1) \ge g(1)$ as claimed.

Step 12. Conclusion: $\displaystyle\int_1^{+\infty} \frac{1 + \cos^2 x}{x^{\frac{2}{3}}}\,dx$ diverges.

Answer. The integral diverges (it is bounded below by $\displaystyle\int_1^{+\infty} \frac{1}{x^{\frac{2}{3}}}\,dx$, a divergent p-integral with $p = \dfrac{2}{3}$).

EXAMPLE 7

Level: Challenge. Basic Comparison Test with a logarithm: a fully justified bound

Determine whether

$$\int_2^{+\infty} \frac{(\ln x)^{10}}{x^2}\,dx$$

converges or diverges.

Step 1. Write $f(x) = \dfrac{(\ln x)^{10}}{x^2}$. For $x \ge 2$ we have $\ln x \ge \ln 2 > 0$ and $x^2 > 0$, so f is continuous and positive on $[2, +\infty)$. There is no elementary antiderivative that is convenient, so we use a comparison.

Step 2. The logarithm grows more slowly than any positive power of x, so we expect $(\ln x)^{10}$ to be dominated by a small power such as $x^{\frac{1}{2}}$, leaving a p-integral with $p = 2 - \dfrac{1}{2} = \dfrac{3}{2} > 1$. We now prove a bound valid for every $x \ge 2$ (no unproved phrase such as 'for large enough x').

Step 3. Lemma: $\ln y \le y - 1$ for every $y > 0$. Proof: let $h(y) = y - 1 - \ln y$. Then

$h'(y) = 1 - \dfrac{1}{y}$, which is negative for $0 < y < 1$ and positive for $y > 1$, so h has its minimum at $y = 1$, where $h(1) = 1 - 1 - 0 = 0$. Hence $h(y) \geq 0$ for all $y > 0$, that is $\ln y \leq y - 1 < y$.

Step 4. Apply the lemma with $y = x^{\frac{1}{20}}$, which is positive. By the logarithm power rule $\ln\left(x^{\frac{1}{20}}\right) = \dfrac{1}{20}\ln x$, so $\dfrac{1}{20}\ln x \leq x^{\frac{1}{20}}$. Multiply by 20:

$$\ln x \leq 20\, x^{\frac{1}{20}} \qquad \text{for all } x > 0.$$

Step 5. Raise both sides to the power 10. For $x \geq 2$ both sides are positive ($0 < \ln x \leq 20\, x^{\frac{1}{20}}$), and $t \mapsto t^{10}$ is increasing on positive numbers, so the inequality is preserved. Using $(ab)^{10} = a^{10}b^{10}$ and $\left(x^{\frac{1}{20}}\right)^{10} = x^{\frac{10}{20}} = x^{\frac{1}{2}}$:

$$(\ln x)^{10} \leq 20^{10}\, x^{\frac{1}{2}} \qquad \text{for all } x \geq 2.$$

Step 6. Divide by $x^2 > 0$ and use the exponent rule $\dfrac{x^{\frac{1}{2}}}{x^2} = x^{\frac{1}{2}-2} = x^{-\frac{3}{2}}$:

$$0 < f(x) = \frac{(\ln x)^{10}}{x^2} \leq \frac{20^{10}}{x^{\frac{3}{2}}} = g(x) \qquad \text{for all } x \geq 2.$$

The bound holds on the entire interval, so no 'eventually' argument is needed. The constant 20^{10} is huge, but a constant factor does not affect convergence.

Step 7. Now compute the comparison integral from the definition. By the power rule with $n = -\dfrac{3}{2}$, $n + 1 = -\dfrac{1}{2}$:

$$\int_2^t x^{-\frac{3}{2}}\, dx = \left[\frac{x^{-\frac{1}{2}}}{-\frac{1}{2}}\right]_2^t = \left[-\frac{2}{\sqrt{x}}\right]_2^t = -\frac{2}{\sqrt{t}} + \frac{2}{\sqrt{2}} = \sqrt{2} - \frac{2}{\sqrt{t}}.$$

Step 8. Check the antiderivative: $\dfrac{d}{dx}\left[-2x^{-\frac{1}{2}}\right] = -2 \cdot \left(-\dfrac{1}{2}\right) x^{-\frac{3}{2}} = x^{-\frac{3}{2}}$.

Step 9. Let $t \to +\infty$: $\dfrac{2}{\sqrt{t}} \to 0$, so $\displaystyle\int_2^{+\infty} x^{-\frac{3}{2}}\, dx = \sqrt{2}$, which is finite. Therefore $\displaystyle\int_2^{+\infty} g(x)\, dx = 20^{10}\sqrt{2}$ converges (a finite constant times a convergent integral).

Step 10. State the Basic Comparison Test, part (ii): if $0 \leq f(x) \leq g(x)$ on $[a, +\infty)$ and

$\int_a^{+\infty} g(x)\,dx$ converges, then $\int_a^{+\infty} f(x)\,dx$ converges (the bigger function converges, so the smaller one converges).

Step 11. Concretely, $F(t) = \int_2^t f(x)\,dx$ is increasing in t (because $f > 0$) and bounded above by $20^{10}\sqrt{2}$ for every $t \geq 2$. An increasing function of t that is bounded above has a finite limit as $t \to +\infty$ (monotone convergence), so the improper integral converges.

Step 12. Conclusion: $\int_2^{+\infty} \frac{(\ln x)^{10}}{x^2}\,dx$ converges, and its value is at most $20^{10}\sqrt{2}$.

Answer. The integral converges, since $0 < \frac{(\ln x)^{10}}{x^2} \leq \frac{20^{10}}{x^{\frac{3}{2}}}$ for all $x \geq 2$ and $\int_2^{+\infty} \frac{dx}{x^{\frac{3}{2}}} = \sqrt{2}$.

EXAMPLE 8

Level: Advanced. Limit Comparison Test

Determine whether

$$\int_1^{+\infty} \frac{x+2}{\sqrt{x^4+x+1}}\,dx$$

converges or diverges.

Step 1. Write $f(x) = \frac{x+2}{\sqrt{x^4+x+1}}$. For $x \geq 1$ the numerator is positive and the radicand $x^4 + x + 1 \geq 3$ is positive, so f is continuous and positive on $[1, +\infty)$.

Step 2. A pointwise inequality such as $f(x) \leq \frac{1}{x}$ is awkward to prove here because of the square root, so use the Limit Comparison Test. First find the benchmark by keeping only the dominant term in each part for large x: the numerator behaves like x, and $\sqrt{x^4+x+1}$ behaves like $\sqrt{x^4} = x^2$.

Step 3. So $f(x)$ behaves like $\frac{x}{x^2} = \frac{1}{x}$. Choose $g(x) = \frac{1}{x}$, which is continuous and positive on $[1, +\infty)$.

Step 4. Form the ratio and simplify the complex fraction by multiplying by the reciprocal of $\frac{1}{x}$, which is x:

$$\frac{f(x)}{g(x)} = \frac{x+2}{\sqrt{x^4+x+1}} \cdot x = \frac{x^2+2x}{\sqrt{x^4+x+1}}.$$

Step 5. Factor the highest power out of the numerator: $x^2 + 2x = x^2\left(1 + \frac{2}{x}\right)$.

Step 6. Factor the highest power out of the radicand: $x^4 + x + 1 = x^4\left(1 + \frac{1}{x^3} + \frac{1}{x^4}\right)$. Since $x > 0$, $\sqrt{x^4} = x^2$, so $\sqrt{x^4+x+1} = x^2\sqrt{1 + \frac{1}{x^3} + \frac{1}{x^4}}$.

Step 7. Substitute and cancel the common factor x^2:

$$\frac{f(x)}{g(x)} = \frac{x^2\left(1+\dfrac{2}{x}\right)}{x^2\sqrt{1+\dfrac{1}{x^3}+\dfrac{1}{x^4}}} = \frac{1+\dfrac{2}{x}}{\sqrt{1+\dfrac{1}{x^3}+\dfrac{1}{x^4}}}.$$

Step 8. Take the limit as $x \to +\infty$. Each of $\dfrac{2}{x}$, $\dfrac{1}{x^3}$, $\dfrac{1}{x^4}$ tends to 0, and the square root is continuous at 1, so

$$L = \lim_{x\to+\infty} \frac{f(x)}{g(x)} = \frac{1+0}{\sqrt{1+0+0}} = 1.$$

Step 9. Numerical spot check at $x = 1000$: $f(1000) = \dfrac{1002}{\sqrt{10^{12}+1001}} \approx \dfrac{1002}{1.0000000005\times 10^6} \approx 1.002\times 10^{-3}$, and $g(1000) = 10^{-3}$, so the ratio is about 1.002, close to $L = 1$.

Step 10. State the Limit Comparison Test: if $f, g \geq 0$ are continuous on $[a, +\infty)$ and $\lim\limits_{x\to+\infty} \dfrac{f(x)}{g(x)} = L$ with $0 < L < \infty$, then $\displaystyle\int_a^{+\infty} f(x)\,dx$ and $\displaystyle\int_a^{+\infty} g(x)\,dx$ either both converge or both diverge. Here $L = 1$ is finite and nonzero, so the test applies.

Step 11. Decide the benchmark integral: $\displaystyle\int_1^t \frac{1}{x}\,dx = \ln t - \ln 1 = \ln t \to +\infty$ as $t \to +\infty$. So $\displaystyle\int_1^{+\infty} \frac{1}{x}\,dx$ diverges (it is the p-integral with $p = 1$).

Step 12. Conclusion: since the benchmark diverges and the two integrals behave the same way, $\displaystyle\int_1^{+\infty} \frac{x+2}{\sqrt{x^4+x+1}}\,dx$ diverges.

Answer. The integral diverges (LCT with $g(x) = \dfrac{1}{x}$, $L = 1$).

Practice Problems

Worked-solutions version: every step of each solution is shown in green beneath its problem.

Computing from the definition

1. Evaluate $\displaystyle\int_2^{+\infty} \frac{1}{(x-1)^2}\,dx$, or show it diverges.

SOLUTION

Step 1. Classify the integral. The upper limit is $+\infty$, so it is Type 1. The integrand $\dfrac{1}{(x-1)^2}$ is continuous on $[2, +\infty)$ because $x - 1 \geq 1$ there, so the denominator

never vanishes.

Step 2. Apply the definition:

$$\int_2^{+\infty} \frac{1}{(x-1)^2}\,dx = \lim_{t\to+\infty} \int_2^t \frac{1}{(x-1)^2}\,dx.$$

Step 3. Use the substitution $u = x - 1$ so that the inner expression becomes a simple power. Then $du = dx$.

Step 4. Change the limits of integration. When $x = 2$: $u = 2-1 = 1$. When $x = t$: $u = t - 1$.

Step 5. Rewrite the integral in u:

$$\int_1^{t-1} u^{-2}\,du.$$

Step 6. Apply the power rule $\int u^k\,du = \dfrac{u^{k+1}}{k+1}$ with $k = -2$: $\int u^{-2}\,du = \dfrac{u^{-1}}{-1} = -\dfrac{1}{u}$.

Step 7. Evaluate at the upper limit $u = t - 1$: $-\dfrac{1}{t-1}$. Evaluate at the lower limit $u = 1$: $-\dfrac{1}{1} = -1$.

Step 8. Subtract (upper minus lower):

$$-\frac{1}{t-1} - (-1) = 1 - \frac{1}{t-1}.$$

Step 9. Take the limit. As $t \to +\infty$, $t-1 \to +\infty$, so $\dfrac{1}{t-1} \to 0$ and the expression tends to 1.

Step 10. Check the antiderivative in the original variable: $\dfrac{d}{dx}\left[-\dfrac{1}{x-1}\right] = \dfrac{d}{dx}\left[-(x-1)^{-1}\right] = (x-1)^{-2}$. Numerical check: for $t = 101$ the partial integral is $1 - \dfrac{1}{100} = 0.99$.

Answer. Converges to 1

2. Evaluate $\displaystyle\int_0^{+\infty} x\,e^{-2x}\,dx$, or show it diverges.

SOLUTION

Step 1. Classify the integral. The upper limit is $+\infty$, so it is Type 1:

$$\int_0^{+\infty} xe^{-2x}\,dx = \lim_{t\to+\infty} \int_0^t xe^{-2x}\,dx.$$

The integrand is continuous everywhere.

Step 2. The integrand is a product of a polynomial x and an exponential, so use integration by parts, $\int u\,dv = uv - \int v\,du$. Choose $u = x$ (it becomes simpler when differentiated) and $dv = e^{-2x}\,dx$.

Step 3. Differentiate u: $du = dx$. Integrate dv: with $w = -2x$, $dw = -2\,dx$, so

$\int e^{-2x}\,dx = -\frac{1}{2}\int e^{w}\,dw = -\frac{1}{2}e^{-2x}$. Thus $v = -\frac{1}{2}e^{-2x}$.

Step 4. Apply the parts formula:

$$\int xe^{-2x}\,dx = x\left(-\frac{1}{2}e^{-2x}\right) - \int\left(-\frac{1}{2}e^{-2x}\right)dx = -\frac{1}{2}xe^{-2x} + \frac{1}{2}\int e^{-2x}\,dx.$$

Step 5. Evaluate the remaining integral using the same antiderivative: $\frac{1}{2}\int e^{-2x}\,dx = \frac{1}{2}\left(-\frac{1}{2}e^{-2x}\right) = -\frac{1}{4}e^{-2x}$.

Step 6. So $F(x) = -\frac{1}{2}xe^{-2x} - \frac{1}{4}e^{-2x}$. Check by differentiating with the product rule: $\frac{d}{dx}\left[-\frac{1}{2}xe^{-2x}\right] = -\frac{1}{2}e^{-2x} + xe^{-2x}$ and $\frac{d}{dx}\left[-\frac{1}{4}e^{-2x}\right] = \frac{1}{2}e^{-2x}$. The sum is xe^{-2x}.

Step 7. Evaluate on $[0, t]$. At the lower limit: $F(0) = 0 - \frac{1}{4}e^{0} = -\frac{1}{4}$. At the upper limit: $F(t) = -\frac{1}{2}te^{-2t} - \frac{1}{4}e^{-2t}$. So

$$\int_0^t xe^{-2x}\,dx = F(t) - F(0) = -\frac{1}{2}te^{-2t} - \frac{1}{4}e^{-2t} + \frac{1}{4}.$$

Step 8. Take the limit. First, $e^{-2t} \to 0$ as $t \to +\infty$.

Step 9. Second, $te^{-2t} = \frac{t}{e^{2t}}$ is of the form $\frac{\infty}{\infty}$, so apply L'Hôpital's rule: $\lim_{t\to\infty}\frac{t}{e^{2t}} = \lim_{t\to\infty}\frac{1}{2e^{2t}} = 0$.

Step 10. Substitute both limits: $-\frac{1}{2}\cdot 0 - \frac{1}{4}\cdot 0 + \frac{1}{4} = \frac{1}{4}$. The limit is finite, so the integral converges to $\frac{1}{4}$.

Answer. Converges to $\frac{1}{4}$

3. Evaluate $\displaystyle\int_{-\infty}^{0}\frac{1}{x^2+4}\,dx$, or show it diverges.

SOLUTION

Step 1. Classify the integral. The lower limit is $-\infty$, so it is Type 1:

$$\int_{-\infty}^{0}\frac{1}{x^2+4}\,dx = \lim_{t\to-\infty}\int_t^0\frac{1}{x^2+4}\,dx.$$

The integrand is continuous everywhere since $x^2 + 4 \geq 4$.

Step 2. The denominator has the form $x^2 + a^2$ with $a = 2$, which points to an arctangent. Derive the antiderivative with the substitution $x = 2u$, so $dx = 2\,du$.

Step 3. Substitute: $\frac{1}{x^2+4} = \frac{1}{4u^2+4} = \frac{1}{4(u^2+1)}$, so

$$\int \frac{dx}{x^2+4} = \int \frac{2\,du}{4(u^2+1)} = \frac{1}{2}\int \frac{du}{u^2+1}.$$

Step 4. Use the standard integral $\int \frac{du}{1+u^2} = \arctan u$ to get $\frac{1}{2}\arctan u$, and replace $u = \frac{x}{2}$: the antiderivative is $F(x) = \frac{1}{2}\arctan\frac{x}{2}$.

Step 5. Check by the chain rule: $F'(x) = \frac{1}{2}\cdot\frac{1}{1+\frac{x^2}{4}}\cdot\frac{1}{2} = \frac{1}{4}\cdot\frac{4}{4+x^2} = \frac{1}{x^2+4}$.

Step 6. Evaluate on $[t,0]$: $F(0) = \frac{1}{2}\arctan 0 = 0$ and $F(t) = \frac{1}{2}\arctan\frac{t}{2}$, so

$$\int_t^0 \frac{dx}{x^2+4} = 0 - \frac{1}{2}\arctan\frac{t}{2} = -\frac{1}{2}\arctan\frac{t}{2}.$$

Step 7. Take the limit $t \to -\infty$. Then $\frac{t}{2} \to -\infty$, and arctan has the horizontal asymptote $-\frac{\pi}{2}$ on the left, so $\arctan\frac{t}{2} \to -\frac{\pi}{2}$.

Step 8. Therefore the limit is $-\frac{1}{2}\left(-\frac{\pi}{2}\right) = \frac{\pi}{4}$, which is finite, so the integral converges.

Step 9. Numerical check: for $t = -100$, $-\frac{1}{2}\arctan(-50) = \frac{1}{2}(1.5508) = 0.7754$, close to $\frac{\pi}{4} \approx 0.7854$.

Answer. Converges to $\frac{\pi}{4}$

Type-2 and split integrals

4. Evaluate $\displaystyle\int_0^1 \frac{1}{\sqrt{x}}\,dx$, or show it diverges.

SOLUTION

Step 1. Locate the discontinuity. The integrand $\frac{1}{\sqrt{x}}$ is continuous for $x > 0$, but at $x = 0$ the denominator vanishes and the function is unbounded. The bad point is the lower endpoint, so this is Type 2.

Step 2. By definition, approach the lower endpoint from the right:

$$\int_0^1 \frac{1}{\sqrt{x}}\,dx = \lim_{t\to 0^+}\int_t^1 \frac{1}{\sqrt{x}}\,dx.$$

Step 3. Rewrite $\frac{1}{\sqrt{x}} = x^{-1/2}$ and apply the power rule with $k = -\frac{1}{2}$, $k+1 = \frac{1}{2}$:

$$\int x^{-1/2}\,dx = \frac{x^{1/2}}{\frac{1}{2}} = 2\sqrt{x}.$$

Step 4. Check: $\frac{d}{dx}\left[2x^{1/2}\right] = 2 \cdot \frac{1}{2}x^{-1/2} = x^{-1/2}$.

Step 5. Evaluate on $[t, 1]$: upper limit $2\sqrt{1} = 2$, lower limit $2\sqrt{t}$. So

$$\int_t^1 x^{-1/2}\,dx = 2 - 2\sqrt{t}.$$

Step 6. Take the limit $t \to 0^+$: $\sqrt{t} \to 0$, so the expression tends to 2.

Step 7. The limit is finite, so the integral converges to 2.

Step 8. Numerical check: for $t = 0.0001$, $2 - 2(0.01) = 1.98$, close to 2.

Answer. Converges to 2

5. Determine whether $\displaystyle\int_{-8}^{8} \frac{1}{x^{2/3}}\,dx$ converges or diverges, and find its value if it converges.

SOLUTION

Step 1. Find all discontinuities in $[-8, 8]$. Write $x^{2/3} = \left(\sqrt[3]{x}\right)^2$, which is defined for negative x as well and is positive for every $x \neq 0$. The denominator is zero only at $x = 0$, so the integrand is unbounded there. Since 0 lies strictly between -8 and 8, the discontinuity is an interior point.

Step 2. An integral with an interior discontinuity must be split at that point, and it converges only if both pieces converge:

$$\int_{-8}^{8} \frac{dx}{x^{2/3}} = \int_{-8}^{0} \frac{dx}{x^{2/3}} + \int_{0}^{8} \frac{dx}{x^{2/3}}.$$

Each piece is Type 2 with the discontinuity at one endpoint.

Step 3. Find an antiderivative. By the power rule with $k = -\frac{2}{3}$, $k + 1 = \frac{1}{3}$: $\int x^{-2/3}\,dx = \frac{x^{1/3}}{\frac{1}{3}} = 3\sqrt[3]{x}$. Check: $\frac{d}{dx}\left[3x^{1/3}\right] = x^{-2/3}$, valid for both positive and negative x (with the real cube root).

Step 4. Right piece: $\displaystyle\int_0^8 = \lim_{t\to 0^+}\left[3\sqrt[3]{x}\right]_t^8 = \lim_{t\to 0^+}\left(3 \cdot 2 - 3\sqrt[3]{t}\right)$, using $\sqrt[3]{8} = 2$.

Step 5. As $t \to 0^+$, $\sqrt[3]{t} \to 0$, so the right piece converges to 6.

Step 6. Left piece: $\displaystyle\int_{-8}^0 = \lim_{s\to 0^-}\left[3\sqrt[3]{x}\right]_{-8}^s = \lim_{s\to 0^-}\left(3\sqrt[3]{s} - 3\sqrt[3]{-8}\right)$, and $\sqrt[3]{-8} = -2$, so this is $3\sqrt[3]{s} + 6$.

Step 7. As $s \to 0^-$, $\sqrt[3]{s} \to 0$, so the left piece converges to 6.

Step 8. Both pieces converge, so the integral converges, with value $6 + 6 = 12$. Consistency check: the integrand $x^{-2/3}$ is even (replacing x by $-x$ does not change

$(\sqrt[3]{x})^2$), so the two halves must be equal, as found.
Answer. Converges to 12 (the discontinuity at $x = 0$ splits it into two halves, each equal to 6)

6. A student evaluates $\displaystyle\int_0^2 \frac{1}{(x-1)^2}\,dx$ as follows: $\displaystyle\int_0^2 (x-1)^{-2}\,dx = \left[-\frac{1}{x-1}\right]_0^2 = -1-1 = -2$. Explain precisely what is wrong with this computation, and determine the correct answer (a convergent value, or divergence).

SOLUTION

Step 1. First test the antiderivative: $\dfrac{d}{dx}\left[-\dfrac{1}{x-1}\right] = \dfrac{d}{dx}\left[-(x-1)^{-1}\right] = (x-1)^{-2}$. So the antiderivative itself is correct, and the error lies in how it was used.

Step 2. Look for discontinuities of the integrand $\dfrac{1}{(x-1)^2}$ on $[0,2]$. The denominator is zero at $x = 1$, and 1 lies inside $[0,2]$, so the integrand is unbounded at an interior point and the integral is improper (Type 2).

Step 3. The Fundamental Theorem of Calculus, $\int_a^b f(x)\,dx = F(b) - F(a)$, requires f to be continuous on all of $[a,b]$ (equivalently, F must be a continuous antiderivative on $[a,b]$). Here $F(x) = -\dfrac{1}{x-1}$ is undefined at $x = 1$ and has a vertical asymptote there, so the theorem does not apply across $[0,2]$.

Step 4. The student's arithmetic reproduces what the formula gives: $F(2) = -\dfrac{1}{1} = -1$ and $F(0) = -\dfrac{1}{-1} = 1$, so $F(2) - F(0) = -1 - 1 = -2$. This number is meaningless, and a quick sign test confirms it: $\dfrac{1}{(x-1)^2} > 0$ for every $x \neq 1$, so no integral of it can be negative.

Step 5. Do it correctly by splitting at the discontinuity:

$$\int_0^2 \frac{dx}{(x-1)^2} = \int_0^1 \frac{dx}{(x-1)^2} + \int_1^2 \frac{dx}{(x-1)^2},$$

and the whole integral converges only if both pieces converge.

Step 6. Examine the first piece: $\displaystyle\int_0^1 = \lim_{t\to 1^-}\left[-\frac{1}{x-1}\right]_0^t = \lim_{t\to 1^-}\left(-\frac{1}{t-1} - 1\right)$, because $F(0) = 1$ so $F(t) - F(0) = -\dfrac{1}{t-1} - 1$.

Step 7. As $t \to 1^-$, the quantity $t-1$ is a small negative number approaching 0, so $-\dfrac{1}{t-1}$ is a large positive number: $-\dfrac{1}{t-1} \to +\infty$. Hence the first piece diverges to $+\infty$.

Step 8. Numerical illustration: for $t = 0.99$ the partial integral is $-\dfrac{1}{-0.01} - 1 = 99$, and for $t = 0.9999$ it is 9999.

Step 9. Since one piece diverges, the whole integral diverges. (The second piece also diverges: $\int_s^2 = \Big[-\frac{1}{x-1}\Big]_s^2 = -1 + \frac{1}{s-1} \to +\infty$ as $s \to 1^+$.)

Answer. The Fundamental Theorem was applied across the discontinuity at $x = 1$ inside $[0, 2]$. The integral actually diverges (to $+\infty$)

Rapid p-integral and comparison judgment

7. State whether

$$\int_1^{+\infty} \frac{1}{x^5}\,dx$$

converges or diverges, with one line of justification.

SOLUTION

Step 1. The integrand is $\frac{1}{x^p}$ with $p = 5$, and the interval is $[1, +\infty)$ with the only trouble at the upper limit $+\infty$. So this is a p-integral of Type 1.

Step 2. The p-integral rule for this interval is: $\int_1^{+\infty} \frac{1}{x^p}\,dx$ converges if and only if $p > 1$.

Step 3. Here $p = 5 > 1$, so the integral converges.

Step 4. Confirm from the definition: $\int_1^{+\infty} x^{-5}\,dx = \lim_{t\to+\infty} \int_1^t x^{-5}\,dx$.

Step 5. Power rule $\int x^n\,dx = \frac{x^{n+1}}{n+1}$ with $n = -5$, $n+1 = -4$: $\int x^{-5}\,dx = \frac{x^{-4}}{-4} = -\frac{1}{4x^4}$. Check: $\frac{d}{dx}\left[-\frac{1}{4}x^{-4}\right] = -\frac{1}{4}(-4)x^{-5} = x^{-5}$.

Step 6. Evaluate between 1 and t: $\left[-\frac{1}{4x^4}\right]_1^t = -\frac{1}{4t^4} + \frac{1}{4}$.

Step 7. As $t \to +\infty$, $t^4 \to +\infty$ so $\frac{1}{4t^4} \to 0$, and the limit is $\frac{1}{4}$, a finite number.

Step 8. Conclusion: the integral converges, and its value is $\frac{1}{4}$.

Answer. Converges ($p = 5 > 1$ on $[1, +\infty)$); the value is $\frac{1}{4}$.

8. State whether

$$\int_0^1 \frac{1}{x^5}\,dx$$

converges or diverges, with one line of justification.

SOLUTION

Step 1. The integrand $\dfrac{1}{x^5}$ is undefined at $x = 0$ and tends to $+\infty$ as $x \to 0^+$. The singularity is at the lower endpoint 0, so this is a Type-2 improper integral, and the relevant benchmark is $\displaystyle\int_0^1 \frac{1}{x^p}\,dx$ with $p = 5$.

Step 2. The p-integral rule for the interval $[0, 1]$ is: $\displaystyle\int_0^1 \frac{1}{x^p}\,dx$ converges if and only if $p < 1$.

Step 3. Here $p = 5$, which is not less than 1, so the integral diverges.

Step 4. Confirm from the definition: $\displaystyle\int_0^1 x^{-5}\,dx = \lim_{t\to 0^+} \int_t^1 x^{-5}\,dx.$

Step 5. An antiderivative is $-\dfrac{1}{4x^4}$ (power rule with $n = -5$, $n+1 = -4$; derivative check: $-\dfrac{1}{4}(-4)x^{-5} = x^{-5}$).

Step 6. Evaluate between t and 1: $\displaystyle\left[-\frac{1}{4x^4}\right]_t^1 = -\frac{1}{4} + \frac{1}{4t^4}.$

Step 7. As $t \to 0^+$, $t^4 \to 0^+$, so $\dfrac{1}{4t^4} \to +\infty$ and the whole expression tends to $+\infty$.

Step 8. Conclusion: the integral diverges to $+\infty$.

Answer. Diverges ($p = 5 \geq 1$ on $[0, 1]$); $\displaystyle\int_t^1 x^{-5}\,dx = \frac{1}{4t^4} - \frac{1}{4} \to +\infty$ as $t \to 0^+$.

9. Use the Basic Comparison Test to decide whether

$$\int_1^{+\infty} \frac{2 + \sin x}{x^{\frac{3}{4}}}\,dx$$

converges or diverges.

SOLUTION

Step 1. Let $f(x) = \dfrac{2 + \sin x}{x^{\frac{3}{4}}}$. The sine takes values in $[-1, 1]$, so $2 + \sin x \geq 2 - 1 = 1 > 0$, and $x^{\frac{3}{4}} > 0$ for $x \geq 1$. So f is continuous and positive on $[1, +\infty)$.

Step 2. Bound the numerator from below: $\sin x \geq -1$ for all x, so $2 + \sin x \geq 1$.

Step 3. Divide by $x^{\frac{3}{4}} > 0$ (the inequality direction is kept):

$$f(x) \geq \frac{1}{x^{\frac{3}{4}}} = g(x) > 0 \qquad \text{for all } x \geq 1.$$

Step 4. The benchmark $\displaystyle\int_1^{+\infty} \frac{1}{x^{\frac{3}{4}}}\,dx$ is a p-integral with $p = \frac{3}{4} \leq 1$, so it should diverge. Confirm: power rule with $n = -\frac{3}{4}$, $n+1 = \frac{1}{4}$ gives $\int x^{-\frac{3}{4}}\,dx = 4x^{\frac{1}{4}}$ (check: $\frac{d}{dx}\left[4x^{\frac{1}{4}}\right] = x^{-\frac{3}{4}}$).

Step 5. Evaluate: $\displaystyle\int_1^t x^{-\frac{3}{4}}\,dx = \left[4x^{\frac{1}{4}}\right]_1^t = 4t^{\frac{1}{4}} - 4 \to +\infty$ as $t \to +\infty$. So the smaller function g has a divergent integral.

Step 6. Basic Comparison Test, part (i): if $0 \leq g \leq f$ and $\displaystyle\int_1^{+\infty} g = +\infty$, then $\displaystyle\int_1^{+\infty} f = +\infty$. Here g is the smaller function and it diverges, so f diverges.

Step 7. Written out, $\displaystyle\int_1^t f(x)\,dx \geq 4t^{\frac{1}{4}} - 4$ for every $t \geq 1$, and the right side tends to $+\infty$.

Step 8. Conclusion: the integral diverges.

Answer. Diverges, because $\dfrac{2+\sin x}{x^{\frac{3}{4}}} \geq \dfrac{1}{x^{\frac{3}{4}}}$ and $\displaystyle\int_1^{+\infty} \frac{dx}{x^{\frac{3}{4}}}$ diverges ($p = \frac{3}{4} \leq 1$).

10. Determine whether

$$\int_0^{+\infty} \frac{1}{x^{\frac{3}{2}}}\,dx$$

converges or diverges.

SOLUTION

Step 1. The integrand $\dfrac{1}{x^{\frac{3}{2}}}$ has two sources of impropriety: it blows up as $x \to 0^+$ (Type 2 at the lower limit), and the upper limit is $+\infty$ (Type 1). One limit cannot handle both.

Step 2. By the definition for mixed integrals, split at any interior point; choose 1:

$$\int_0^{+\infty} x^{-\frac{3}{2}}\,dx = \int_0^1 x^{-\frac{3}{2}}\,dx + \int_1^{+\infty} x^{-\frac{3}{2}}\,dx.$$

The whole integral converges only if both pieces converge.

Step 3. An antiderivative, by the power rule with $n = -\frac{3}{2}$, $n+1 = -\frac{1}{2}$: $\int x^{-\frac{3}{2}}\,dx = \frac{x^{-\frac{1}{2}}}{-\frac{1}{2}} = -\frac{2}{\sqrt{x}}$. Check: $\frac{d}{dx}\left[-2x^{-\frac{1}{2}}\right] = -2\left(-\frac{1}{2}\right)x^{-\frac{3}{2}} = x^{-\frac{3}{2}}$.

Step 4. First piece: $\displaystyle\int_0^1 x^{-\frac{3}{2}}\,dx = \lim_{t\to 0^+}\left[-\frac{2}{\sqrt{x}}\right]_t^1 = \lim_{t\to 0^+}\left(-2+\frac{2}{\sqrt{t}}\right)$.

Step 5. As $t \to 0^+$, $\sqrt{t} \to 0^+$, so $\frac{2}{\sqrt{t}} \to +\infty$. The first piece diverges to $+\infty$. (This agrees with the rule that $\displaystyle\int_0^1 \frac{dx}{x^p}$ needs $p < 1$, while here $p = \frac{3}{2}$.)

Step 6. Because one piece diverges, the whole integral diverges, whatever the other piece does.

Step 7. For completeness, second piece: $\displaystyle\int_1^{+\infty} x^{-\frac{3}{2}}\,dx = \lim_{T\to+\infty}\left[-\frac{2}{\sqrt{x}}\right]_1^T = \lim_{T\to+\infty}\left(-\frac{2}{\sqrt{T}}+2\right) = 2$, which converges.

Step 8. The second piece converging cannot repair the first. In general $\displaystyle\int_0^{+\infty}\frac{dx}{x^p}$ needs $p < 1$ for the piece on $[0,1]$ and $p > 1$ for the piece on $[1,+\infty)$, which cannot both hold.

Step 9. Conclusion: the integral diverges.

Answer. Diverges: $\displaystyle\int_0^1 x^{-\frac{3}{2}}\,dx = +\infty$ (even though $\displaystyle\int_1^{+\infty} x^{-\frac{3}{2}}\,dx = 2$).

BCT / LCT applications

11. Use the Basic Comparison Test to determine whether

$$\int_1^{+\infty} \frac{\arctan\left(x^2\right)}{1+x^3}\,dx$$

converges or diverges.

SOLUTION

Step 1. Let $f(x) = \dfrac{\arctan\left(x^2\right)}{1+x^3}$. For $x \geq 1$, $x^2 > 0$ so $\arctan\left(x^2\right) > 0$, and $1 + x^3 > 0$. So f is continuous and positive on $[1, +\infty)$.

Step 2. Bound the numerator from above. The range of arctan is $\left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$, so $\arctan\left(x^2\right) < \dfrac{\pi}{2}$ for every x.

Step 3. Bound the denominator from below: $1 + x^3 > x^3 > 0$. Taking reciprocals of positive numbers reverses the inequality: $\dfrac{1}{1+x^3} < \dfrac{1}{x^3}$.

Step 4. Multiply the two inequalities (all quantities are positive):

$$0 < f(x) = \arctan\left(x^2\right) \cdot \frac{1}{1+x^3} < \frac{\pi}{2} \cdot \frac{1}{x^3} = g(x) \qquad \text{for all } x \geq 1.$$

Step 5. Spot check at $x = 1$: $f(1) = \dfrac{\arctan 1}{2} = \dfrac{\pi}{8} \approx 0.393$ and $g(1) = \dfrac{\pi}{2} \approx 1.571$, so $f(1) < g(1)$.

Step 6. Benchmark: $\displaystyle\int_1^t \frac{\pi}{2} x^{-3}\, dx = \frac{\pi}{2}\left[-\frac{1}{2x^2}\right]_1^t = \frac{\pi}{2}\left(\frac{1}{2} - \frac{1}{2t^2}\right)$ (power rule with $n = -3$, $n + 1 = -2$).

Step 7. As $t \to +\infty$ this tends to $\dfrac{\pi}{2} \cdot \dfrac{1}{2} = \dfrac{\pi}{4}$, so $\displaystyle\int_1^{+\infty} g(x)\, dx$ converges (it is $\dfrac{\pi}{2}$ times a p-integral with $p = 3 > 1$).

Step 8. Basic Comparison Test, part (ii): since $0 \leq f \leq g$ and $\displaystyle\int_1^{+\infty} g$ converges, $\displaystyle\int_1^{+\infty} f$ converges, to a value at most $\dfrac{\pi}{4}$.

Step 9. Conclusion: the integral converges.

Answer. Converges, since $0 < \dfrac{\arctan\left(x^2\right)}{1+x^3} < \dfrac{\pi}{2} \cdot \dfrac{1}{x^3}$ and $\displaystyle\int_1^{+\infty} \frac{\pi}{2x^3}\, dx = \frac{\pi}{4}$.

12. Use the Limit Comparison Test to determine whether

$$\int_1^{+\infty} \frac{x^3 + 2x + 7}{x^5 + 11x^4 + 1}\, dx$$

converges or diverges.

SOLUTION

Step 1. Let $f(x) = \dfrac{x^3 + 2x + 7}{x^5 + 11x^4 + 1}$. For $x \geq 1$ every term in the numerator and denominator is positive, so f is continuous and positive on $[1, +\infty)$.

Step 2. Find the benchmark from the leading terms: the numerator behaves like

x^3 and the denominator like x^5, so $f(x)$ behaves like $\dfrac{x^3}{x^5} = \dfrac{1}{x^2}$. Choose $g(x) = \dfrac{1}{x^2}$.

Step 3. Form the ratio $\dfrac{f(x)}{g(x)} = f(x) \cdot x^2$ (dividing by $\dfrac{1}{x^2}$ means multiplying by x^2):

$$\frac{f(x)}{g(x)} = \frac{x^2\,(x^3 + 2x + 7)}{x^5 + 11x^4 + 1} = \frac{x^5 + 2x^3 + 7x^2}{x^5 + 11x^4 + 1}.$$

Step 4. Divide the numerator and the denominator by the highest power, x^5:

$$\frac{f(x)}{g(x)} = \frac{1 + \dfrac{2}{x^2} + \dfrac{7}{x^3}}{1 + \dfrac{11}{x} + \dfrac{1}{x^5}}.$$

Step 5. Take the limit as $x \to +\infty$. Every fraction with a power of x in the denominator tends to 0, so

$$L = \frac{1 + 0 + 0}{1 + 0 + 0} = 1.$$

Step 6. Numerical spot check at $x = 1000$: the numerator is $10^9 + 2000 + 7 = 1{,}000{,}002{,}007$ and the denominator is $10^{15} + 11 \times 10^{12} + 1 = 1{,}011{,}000{,}000{,}001$, so $f(1000) \approx 9.89 \times 10^{-7}$, and $f(1000) \cdot 10^6 \approx 0.989$, close to 1.

Step 7. Since $L = 1$ is finite and nonzero, the Limit Comparison Test says $\displaystyle\int_1^{+\infty} f$ and $\displaystyle\int_1^{+\infty} g$ both converge or both diverge.

Step 8. Decide the benchmark: $\displaystyle\int_1^t x^{-2}\,dx = \left[-\frac{1}{x}\right]_1^t = 1 - \frac{1}{t} \to 1$, so $\displaystyle\int_1^{+\infty} \frac{1}{x^2}\,dx = 1$ converges ($p = 2 > 1$).

Step 9. Conclusion: the given integral converges.

Answer. Converges (LCT with $g(x) = \dfrac{1}{x^2}$, $L = 1$, and $\displaystyle\int_1^{+\infty} \frac{dx}{x^2} = 1$).

13. Use the Limit Comparison Test with $g(x) = \dfrac{1}{x}$ to determine whether

$$\int_2^{+\infty} \frac{\ln x}{x + 1}\,dx$$

converges or diverges.

SOLUTION

Step 1. Let $f(x) = \dfrac{\ln x}{x + 1}$. For $x \geq 2$, $\ln x \geq \ln 2 > 0$ and $x + 1 > 0$, so f is

continuous and positive on $[2, +\infty)$; and $g(x) = \dfrac{1}{x}$ is continuous and positive there as well.

Step 2. Form the ratio, multiplying by the reciprocal of g:

$$\frac{f(x)}{g(x)} = \frac{\ln x}{x+1} \cdot x = \ln x \cdot \frac{x}{x+1}.$$

Step 3. Study the two factors separately as $x \to +\infty$. First, $\ln x \to +\infty$. Second,

$$\frac{x}{x+1} = \frac{1}{1+\dfrac{1}{x}} \to \frac{1}{1+0} = 1.$$

Step 4. A factor tending to $+\infty$ times a factor tending to 1 tends to $+\infty$. To make this exact: for $x \geq 1$ we have $x + 1 \leq 2x$, so $\dfrac{x}{x+1} \geq \dfrac{1}{2}$, and therefore $\dfrac{f(x)}{g(x)} \geq \dfrac{1}{2} \ln x \to +\infty$. So $L = +\infty$.

Step 5. Numerical spot check at $x = 10^6$: $\ln\left(10^6\right) \approx 13.82$ and $\dfrac{x}{x+1} \approx 0.999999$, so the ratio is about 13.8, already large and still increasing.

Step 6. State the third case of the Limit Comparison Test: if $\lim\limits_{x \to +\infty} \dfrac{f(x)}{g(x)} = +\infty$ and $\displaystyle\int_a^{+\infty} g(x)\, dx$ diverges, then $\displaystyle\int_a^{+\infty} f(x)\, dx$ diverges.

Step 7. Check the hypothesis on g: $\displaystyle\int_2^t \frac{1}{x}\, dx = \ln t - \ln 2 \to +\infty$ as $t \to +\infty$, so $\displaystyle\int_2^{+\infty} \frac{1}{x}\, dx$ diverges.

Step 8. Both hypotheses of the case hold, so the conclusion applies.

Step 9. Conclusion: $\displaystyle\int_2^{+\infty} \frac{\ln x}{x+1}\, dx$ diverges.

Answer. Diverges: $\lim\limits_{x \to +\infty} \dfrac{f(x)}{g(x)} = \lim\limits_{x \to +\infty} \dfrac{x \ln x}{x+1} = +\infty$ and $\displaystyle\int_2^{+\infty} \frac{dx}{x}$ diverges.

14. Use the Limit Comparison Test to determine whether

$$\int_0^1 \frac{\sin x}{\dfrac{3}{x^{\dfrac{2}{}}}}\, dx$$

converges or diverges.

SOLUTION

Step 1. Let $f(x) = \dfrac{\sin x}{x^{\frac{3}{2}}}$. On $(0, 1]$ we have $0 < x \le 1 < \pi$, so $\sin x > 0$, and $x^{\frac{3}{2}} > 0$. So f is continuous and positive on $(0, 1]$. At $x = 0$ the denominator is 0, so the integral is Type 2 with the singularity at the lower endpoint.

Step 2. Find the benchmark by studying the behaviour near $x = 0$: since $\sin x \approx x$ for small x, $f(x) \approx \dfrac{x}{x^{\frac{3}{2}}} = x^{-\frac{1}{2}}$. Choose $g(x) = \dfrac{1}{x^{\frac{1}{2}}}$, which is continuous and positive on $(0, 1]$ with the same singularity.

Step 3. The Limit Comparison Test works at a Type-2 endpoint exactly as at infinity, but the limit is taken where the singularity is, here $x \to 0^+$.

Step 4. Form the ratio:

$$\frac{f(x)}{g(x)} = \frac{\sin x}{x^{\frac{3}{2}}} \cdot x^{\frac{1}{2}} = \frac{\sin x}{x^{\frac{3}{2} - \frac{1}{2}}} = \frac{\sin x}{x}.$$

Step 5. Use the standard limit $\lim\limits_{x \to 0} \dfrac{\sin x}{x} = 1$. So $L = \lim\limits_{x \to 0^+} \dfrac{f(x)}{g(x)} = 1$, which is finite and nonzero.

Step 6. By the Limit Comparison Test, $\displaystyle\int_0^1 f(x)\,dx$ and $\displaystyle\int_0^1 g(x)\,dx$ both converge or both diverge.

Step 7. Decide the benchmark from the definition: $\displaystyle\int_0^1 x^{-\frac{1}{2}}\,dx = \lim_{t \to 0^+} \left[2\sqrt{x}\right]_t^1 = \lim\limits_{t \to 0^+} \left(2 - 2\sqrt{t}\right) = 2$. Check: $\dfrac{d}{dx}\left[2x^{\frac{1}{2}}\right] = x^{-\frac{1}{2}}$.

Step 8. So the benchmark converges, consistent with the rule that $\displaystyle\int_0^1 \frac{dx}{x^p}$ converges for $p < 1$, here $p = \dfrac{1}{2}$.

Step 9. Conclusion: $\displaystyle\int_0^1 \frac{\sin x}{x^{\frac{3}{2}}}\,dx$ converges.

Answer. Converges (LCT at $x \to 0^+$ with $g(x) = \dfrac{1}{\sqrt{x}}$, $L = \lim\limits_{x \to 0^+} \dfrac{\sin x}{x} = 1$, and $\displaystyle\int_0^1 \frac{dx}{\sqrt{x}} = 2$).

15. Use the Limit Comparison Test to determine whether

$$\int_0^1 \frac{1}{\sqrt{1-x^3}}\,dx$$

converges or diverges.

SOLUTION

Step 1. Let $f(x) = \dfrac{1}{\sqrt{1-x^3}}$. For $0 \le x < 1$ we have $x^3 < 1$, so $1 - x^3 > 0$ and f is continuous and positive on $[0, 1)$. At $x = 1$ the denominator is 0 and $f(x) \to +\infty$ as $x \to 1^-$. So the integral is Type 2 with the singularity at the upper endpoint 1.

Step 2. To choose a benchmark with the same singularity, factor the troublesome expression using the difference of cubes $a^3 - b^3 = (a-b)\left(a^2 + ab + b^2\right)$ with $a = 1$, $b = x$:

$$1 - x^3 = (1 - x)\left(1 + x + x^2\right).$$

Step 3. Check by expanding: $(1-x)\left(1 + x + x^2\right) = 1 + x + x^2 - x - x^2 - x^3 = 1 - x^3$.

Step 4. The factor $(1-x)$ is the one that vanishes at $x = 1$, while $1 + x + x^2$ tends to 3. So near $x = 1$, $f(x) \approx \dfrac{1}{\sqrt{3}\sqrt{1-x}}$. Choose $g(x) = \dfrac{1}{\sqrt{1-x}}$, continuous and positive on $[0, 1)$.

Step 5. Form the ratio, valid for $0 \le x < 1$:

$$\frac{f(x)}{g(x)} = \frac{\sqrt{1-x}}{\sqrt{1-x^3}} = \sqrt{\frac{1-x}{(1-x)\left(1+x+x^2\right)}} = \frac{1}{\sqrt{1+x+x^2}}.$$

Step 6. Take the limit as $x \to 1^-$ (the location of the singularity): $L = \dfrac{1}{\sqrt{1+1+1}} = \dfrac{1}{\sqrt{3}}$, which is finite and nonzero.

Step 7. By the Limit Comparison Test, $\displaystyle\int_0^1 f$ and $\displaystyle\int_0^1 g$ both converge or both diverge.

Step 8. Decide the benchmark from the definition: $\displaystyle\int_0^1 \frac{dx}{\sqrt{1-x}} = \lim_{t \to 1^-} \int_0^t (1 - x)^{-\frac{1}{2}}\,dx$.

Step 9. Antiderivative: with $u = 1 - x$, $du = -dx$, $\displaystyle\int (1-x)^{-\frac{1}{2}}\,dx = -\int u^{-\frac{1}{2}}\,du = -2\sqrt{u} = -2\sqrt{1-x}$. Check: $\dfrac{d}{dx}\left[-2(1-x)^{\frac{1}{2}}\right] = -2 \cdot \dfrac{1}{2}(1-x)^{-\frac{1}{2}} \cdot (-1) = (1-x)^{-\frac{1}{2}}$.

Step 10. Evaluate: $\left[-2\sqrt{1-x}\right]_0^t = -2\sqrt{1-t} + 2 \to 2$ as $t \to 1^-$. So the benchmark converges.

Step 11. Conclusion: $\displaystyle\int_0^1 \frac{1}{\sqrt{1-x^3}}\,dx$ converges.

Answer. Converges (LCT at $x \to 1^-$ with $g(x) = \dfrac{1}{\sqrt{1-x}}$, $L = \dfrac{1}{\sqrt{3}}$, and $\displaystyle\int_0^1 \frac{dx}{\sqrt{1-x}} = 2$).

Applications

16. A random variable X has probability density $f(x) = \dfrac{k}{x^4}$ for $x \geq 2$ and $f(x) = 0$ for $x < 2$. Find the constant k that makes f a valid density, and then find $P(X > 4)$.

SOLUTION

Step 1. A probability density must satisfy $\displaystyle\int_{-\infty}^{+\infty} f(x)\,dx = 1$. Since $f(x) = 0$ for $x < 2$, the part over $(-\infty, 2)$ contributes 0, and the condition becomes $\displaystyle\int_2^{+\infty} \frac{k}{x^4}\,dx = 1$.

Step 2. Pull the constant out and write the improper integral as a limit:

$$k\int_2^{+\infty} x^{-4}\,dx = k\lim_{t\to+\infty}\int_2^t x^{-4}\,dx.$$

Step 3. Apply the power rule with $k_0 = -4$ (so $k_0 + 1 = -3$): $\displaystyle\int x^{-4}\,dx = \frac{x^{-3}}{-3} = -\frac{1}{3x^3}$. Check: $\dfrac{d}{dx}\left[-\dfrac{1}{3}x^{-3}\right] = x^{-4}$.

Step 4. Evaluate on $[2, t]$:

$$\left[-\frac{1}{3x^3}\right]_2^t = -\frac{1}{3t^3} + \frac{1}{3\cdot 8} = \frac{1}{24} - \frac{1}{3t^3}.$$

Step 5. As $t \to +\infty$, $\dfrac{1}{3t^3} \to 0$, so $\displaystyle\int_2^{+\infty} x^{-4}\,dx = \frac{1}{24}$. The integral converges, so a normalisation constant exists.

Step 6. Impose the condition: $k \cdot \dfrac{1}{24} = 1$, so $k = 24$.

Step 7. For the probability, integrate the density over the event $X > 4$:

$$P(X > 4) = \int_4^{+\infty} \frac{24}{x^4}\,dx = 24\lim_{t\to+\infty}\left[-\frac{1}{3x^3}\right]_4^t.$$

Step 8. Evaluate: at $x = 4$, $3\cdot 4^3 = 3\cdot 64 = 192$, so $\left[-\dfrac{1}{3x^3}\right]_4^t = -\dfrac{1}{3t^3} + \dfrac{1}{192} \to \dfrac{1}{192}$.

Step 9. Multiply by 24: $P(X > 4) = \dfrac{24}{192} = \dfrac{1}{8}$.

Step 10. Check with the general formula: for $x_0 \geq 2$, $P(X > x_0) = 24 \cdot \dfrac{1}{3x_0^3} = \dfrac{8}{x_0^3} = \left(\dfrac{2}{x_0}\right)^3$. At $x_0 = 2$ this gives 1 (all the probability), and at $x_0 = 4$ it gives $\left(\dfrac{1}{2}\right)^3 = \dfrac{1}{8}$.

Answer. $k = 24$ and $P(X > 4) = \dfrac{1}{8}$

17. The gravitational force on a satellite of mass m at distance r from the centre of the Earth (mass M) is $F(r) = \dfrac{GMm}{r^2}$, where G is the gravitational constant. Find the work needed to move the satellite from the distance $r = R$ (with $R > 0$) to arbitrarily far away.

SOLUTION

Step 1. Work done by a variable force over a distance is the integral of the force: moving from $r = R$ to $r = b$ requires $W(b) = \displaystyle\int_R^b F(r)\,dr$. Moving "arbitrarily far away" means we let $b \to +\infty$, so the total work is the Type 1 improper integral

$$W = \lim_{b\to+\infty} \int_R^b \frac{GMm}{r^2}\,dr.$$

Step 2. The quantities G, M, m are constants, so pull them out: $W(b) = GMm \displaystyle\int_R^b r^{-2}\,dr$.

Step 3. Apply the power rule with exponent -2: $\int r^{-2}\,dr = \dfrac{r^{-1}}{-1} = -\dfrac{1}{r}$. Check: $\dfrac{d}{dr}\left[-r^{-1}\right] = r^{-2}$.

Step 4. Evaluate between R and b:

$$\left[-\frac{1}{r}\right]_R^b = -\frac{1}{b} - \left(-\frac{1}{R}\right) = \frac{1}{R} - \frac{1}{b}.$$

Step 5. Hence $W(b) = GMm\left(\dfrac{1}{R} - \dfrac{1}{b}\right)$.

Step 6. Take the limit $b \to +\infty$: $\dfrac{1}{b} \to 0$, so $W = GMm\left(\dfrac{1}{R} - 0\right)$.

Step 7. The result is $W = \dfrac{GMm}{R}$. The integral converges, so a finite amount of work suffices to remove the satellite to infinite distance; this is consistent with the p-integral rule, because the force decays like r^{-2} with $p = 2 > 1$.

Step 8. Unit check: GMm/r^2 is a force in newtons and dr is in metres, so the

work is in newton-metres (joules); and $\dfrac{GMm}{R}$ has units of force $\times$ length since $\dfrac{GMm}{R} = \dfrac{GMm}{R^2} \cdot R$.

Step 9. Numerical check with $GMm = 1$ and $R = 1$: $W(10) = 1 - \dfrac{1}{10} = 0.9$ and $W(100) = 0.99$, approaching $1 = \dfrac{GMm}{R}$.

Answer. $W = \dfrac{GMm}{R}$

18. A damped oscillator moves along a line with velocity $v(t) = e^{-2t}\sin(3t)$ metres per second for $t \geq 0$. Find its total displacement $\displaystyle\int_0^{+\infty} e^{-2t}\sin(3t)\,dt$.

SOLUTION

Step 1. This is a Type 1 integral: $I = \lim_{T\to+\infty} I(T)$ with $I(T) = \displaystyle\int_0^T e^{-2t}\sin 3t\,dt$. The integrand is a product of an exponential and a sine, so integrate by parts twice; the original integral will reappear and can be solved for algebraically.

Step 2. First parts: $u = \sin 3t$, $dv = e^{-2t}\,dt$. Then $du = 3\cos 3t\,dt$ and $v = -\dfrac{1}{2}e^{-2t}$. The formula $\int u\,dv = uv - \int v\,du$ gives

$$I(T) = \Big[-\frac{1}{2}e^{-2t}\sin 3t\Big]_0^T + \frac{3}{2}\int_0^T e^{-2t}\cos 3t\,dt.$$

The boundary term equals $-\dfrac{1}{2}e^{-2T}\sin 3T$, since at $t = 0$ it is $-\dfrac{1}{2}\cdot 1\cdot\sin 0 = 0$.

Step 3. Name the new integral $J(T) = \displaystyle\int_0^T e^{-2t}\cos 3t\,dt$, so $I(T) = -\dfrac{1}{2}e^{-2T}\sin 3T + \dfrac{3}{2}J(T)$.

Step 4. Second parts on $J(T)$: $u = \cos 3t$, $dv = e^{-2t}\,dt$, so $du = -3\sin 3t\,dt$ and $v = -\dfrac{1}{2}e^{-2t}$. Then

$$J(T) = \Big[-\frac{1}{2}e^{-2t}\cos 3t\Big]_0^T - \int_0^T \Big(-\frac{1}{2}e^{-2t}\Big)(-3\sin 3t)\,dt = \Big[-\frac{1}{2}e^{-2t}\cos 3t\Big]_0^T - \frac{3}{2}I(T).$$

Step 5. Evaluate the boundary term: at $t = T$ it is $-\dfrac{1}{2}e^{-2T}\cos 3T$; at $t = 0$ it is $-\dfrac{1}{2}\cdot 1\cdot\cos 0 = -\dfrac{1}{2}$. Subtracting, the boundary term is $\dfrac{1}{2} - \dfrac{1}{2}e^{-2T}\cos 3T$. So $J(T) = \dfrac{1}{2} - \dfrac{1}{2}e^{-2T}\cos 3T - \dfrac{3}{2}I(T)$.

Step 6. Substitute $J(T)$ into the equation of step 3:

$$I(T) = -\frac{1}{2}e^{-2T}\sin 3T + \frac{3}{4} - \frac{3}{4}e^{-2T}\cos 3T - \frac{9}{4}I(T).$$

Step 7. Collect the $I(T)$ terms on the left: $I(T) + \frac{9}{4}I(T) = \frac{13}{4}I(T)$. So

$$\frac{13}{4}I(T) = \frac{3}{4} - e^{-2T}\left(\frac{1}{2}\sin 3T + \frac{3}{4}\cos 3T\right).$$

Step 8. Multiply by $\frac{4}{13}$:

$$I(T) = \frac{3}{13} - \frac{e^{-2T}\,(2\sin 3T + 3\cos 3T)}{13}.$$

Step 9. Take the limit $T \to +\infty$. Because $|2\sin 3T + 3\cos 3T| \le 2 + 3 = 5$ and $e^{-2T} \to 0$, the product satisfies $\left|e^{-2T}(2\sin 3T + 3\cos 3T)\right| \le 5e^{-2T} \to 0$, so it tends to 0 by the squeeze theorem.

Step 10. Therefore $I = \frac{3}{13} - 0 = \frac{3}{13}$, a finite number, so the integral converges.

Step 11. Verification by differentiation: let $F(t) = -\dfrac{e^{-2t}(2\sin 3t + 3\cos 3t)}{13}$. Then $F'(t) = \dfrac{2e^{-2t}(2\sin 3t + 3\cos 3t) - e^{-2t}(6\cos 3t - 9\sin 3t)}{13} = \dfrac{e^{-2t}(4\sin 3t + 6\cos 3t - 6\cos 3t + 9\sin 3t)}{13} = e^{-2t}\sin 3t$, and $F(0) = -\frac{3}{13}$, so $F(T) - F(0)$ agrees with $I(T)$.

Answer. Total displacement $= \frac{3}{13}$ metres

19. The lifetime X (in thousands of hours) of a certain component has probability density function

$$f(x) = \begin{cases} \dfrac{k}{x^3}, & x \ge 2, \\ 0, & x < 2. \end{cases}$$

(a) Find the constant k. (b) Find $P(X > 4)$.

SOLUTION

Step 1. For f to be a probability density function, it must be nonnegative and its total integral over the real line must equal 1: $\displaystyle\int_{-\infty}^{+\infty} f(x)\,dx = 1$. Nonnegativity holds as long as $k > 0$.

Step 2. Since $f(x) = 0$ for $x < 2$, that part contributes nothing, and the condition becomes $\displaystyle\int_2^{+\infty} \frac{k}{x^3}\,dx = 1$. This is a Type-1 improper integral, defined as a limit.

Step 3. Pull out the constant: $k \lim\limits_{t\to+\infty} \displaystyle\int_2^t x^{-3}\,dx$. By the power rule with $n = -3$,

$n+1=-2$, an antiderivative is $-\dfrac{1}{2x^2}$. Check: $\dfrac{d}{dx}\left[-\dfrac{1}{2}x^{-2}\right]=x^{-3}$.

Step 4. Evaluate: $\left[-\dfrac{1}{2x^2}\right]_2^t=-\dfrac{1}{2t^2}+\dfrac{1}{2\cdot 4}=\dfrac{1}{8}-\dfrac{1}{2t^2}$.

Step 5. As $t\to+\infty$, $\dfrac{1}{2t^2}\to 0$, so $\displaystyle\int_2^{+\infty}x^{-3}\,dx=\frac{1}{8}$.

Step 6. The normalisation condition becomes $k\cdot\dfrac{1}{8}=1$, so $k=8$.

Step 7. Part (b): $P(X>4)=\displaystyle\int_4^{+\infty}\frac{8}{x^3}\,dx=8\lim_{t\to+\infty}\left[-\frac{1}{2x^2}\right]_4^t=8\lim_{t\to+\infty}\left(\frac{1}{2\cdot 16}-\frac{1}{2t^2}\right)$.

Step 8. Evaluate the limit: $\dfrac{1}{32}-0=\dfrac{1}{32}$, so $P(X>4)=8\cdot\dfrac{1}{32}=\dfrac{1}{4}$.

Step 9. Consistency check: the same computation with lower limit $a\geq 2$ gives $P(X>a)=8\cdot\dfrac{1}{2a^2}=\dfrac{4}{a^2}$. At $a=2$ this equals 1 (the lifetime is certainly above 2), and at $a=4$ it equals $\dfrac{4}{16}=\dfrac{1}{4}$, in agreement.

Answer. (a) $k=8$. (b) $P(X>4)=\dfrac{1}{4}$.

20. The integral $\Gamma(3)=\displaystyle\int_0^{+\infty}x^2e^{-x}\,dx$ appears in the gamma function, which extends the factorial to non-integers. Evaluate it, or show that it diverges.

SOLUTION

Step 1. The upper limit is $+\infty$ and the integrand is continuous on $[0,+\infty)$, so by the definition

$$\int_0^{+\infty}x^2e^{-x}\,dx=\lim_{t\to+\infty}\int_0^t x^2e^{-x}\,dx.$$

Step 2. A polynomial times an exponential calls for integration by parts, $\int u\,dv=uv-\int v\,du$. Choose $u=x^2$ (it becomes simpler on differentiating) and $dv=e^{-x}\,dx$. Then $du=2x\,dx$ and $v=-e^{-x}$.

Step 3. Apply the formula: $\int x^2e^{-x}\,dx=-x^2e^{-x}-\int(-e^{-x})(2x)\,dx=-x^2e^{-x}+2\int xe^{-x}\,dx$.

Step 4. The remaining integral $\int xe^{-x}\,dx$ needs parts again. Choose $u=x$, $dv=e^{-x}\,dx$, so $du=dx$, $v=-e^{-x}$: $\int xe^{-x}\,dx=-xe^{-x}+\int e^{-x}\,dx=-xe^{-x}-e^{-x}$.

Step 5. Substitute back: $\int x^2e^{-x}\,dx=-x^2e^{-x}+2(-xe^{-x}-e^{-x})=-(x^2+2x+2)e^{-x}$.

Step 6. Verify by differentiating with the product rule: $\dfrac{d}{dx}\left[-(x^2+2x+2)e^{-x}\right]=$

$-(2x+2)e^{-x} + (x^2+2x+2)\,e^{-x} = x^2e^{-x}$, the integrand.

Step 7. Evaluate the definite integral from 0 to t: $F(t) - F(0) = -(t^2+2t+2)\,e^{-t} - (-(0+0+2)e^0) = 2 - (t^2+2t+2)\,e^{-t}$.

Step 8. Compute $\lim_{t\to+\infty} t^2e^{-t} = \lim_{t\to+\infty} \dfrac{t^2}{e^t}$. This is of the form $\dfrac{\infty}{\infty}$, so apply L'Hôpital's rule twice: $\dfrac{t^2}{e^t} \to \dfrac{2t}{e^t} \to \dfrac{2}{e^t} \to 0$.

Step 9. Similarly $\lim_{t\to+\infty} \dfrac{t}{e^t} = \lim_{t\to+\infty} \dfrac{1}{e^t} = 0$ and $\lim_{t\to+\infty} e^{-t} = 0$. So $(t^2+2t+2)\,e^{-t} \to 0$.

Step 10. Therefore $\displaystyle\int_0^{+\infty} x^2e^{-x}\,dx = 2 - 0 = 2$, the integral converges.

Step 11. Interpretation: $\Gamma(3) = 2! = 2$, matching the pattern $\Gamma(n+1) = n!$.

Answer. Converges: $\displaystyle\int_0^{+\infty} x^2e^{-x}\,dx = 2$.

Error diagnosis

21. A student claims that $\displaystyle\int_1^{+\infty} \frac{1}{x}\,dx$ converges, because $\dfrac{1}{x} \to 0$ as $x \to +\infty$, so the area under the curve "runs out". Find the error and give the correct result.

SOLUTION

Step 1. State the student's reasoning: "the integrand tends to 0, therefore the improper integral is finite". We test it against the definition rather than against intuition.

Step 2. By definition, the integral converges if and only if the limit of the accumulated area exists and is finite:

$$\int_1^{+\infty} \frac{1}{x}\,dx = \lim_{t\to+\infty} \int_1^t \frac{1}{x}\,dx.$$

Step 3. Compute the accumulated area with the antiderivative $\int \dfrac{1}{x}\,dx = \ln|x|$:

$$\int_1^t \frac{1}{x}\,dx = \ln t - \ln 1 = \ln t.$$

Step 4. Decide the limit. The function $\ln t$ is increasing, and for any bound M we have $\ln t > M$ whenever $t > e^M$ (because $\ln e^M = M$). So $\ln t$ exceeds every bound, and $\lim_{t\to+\infty} \ln t = +\infty$.

Step 5. Therefore the integral diverges to $+\infty$. The student's conclusion is false.

Step 6. Locate the flaw in the logic. The condition $f(x) \to 0$ only says the height of the curve shrinks; convergence depends on whether the heights shrink fast enough for the total area to stay bounded. A comparison shows this: $\dfrac{1}{x^2}$ also tends to 0,

yet $\displaystyle\int_1^{+\infty} \frac{dx}{x^2} = \lim_{t\to\infty}\left(1-\frac{1}{t}\right) = 1$, while $\dfrac{1}{x}$ shrinks too slowly and its area grows without bound.

Step 7. Numerical illustration of the growth: $\ln 10 \approx 2.30$, $\ln 1000 \approx 6.91$, $\ln 10^6 \approx 13.82$. The partial areas keep increasing and never settle.

Answer. The error is treating $\dfrac{1}{x} \to 0$ as sufficient for convergence. In fact $\displaystyle\int_1^t \frac{dx}{x} = \ln t \to +\infty$, so the integral diverges

22. A student claims that $\displaystyle\int_{-\infty}^{+\infty} \frac{x}{x^2+1}\,dx = 0$, reasoning: "the integrand is odd, so $\displaystyle\int_{-t}^{t} \frac{x}{x^2+1}\,dx = 0$ for every t, hence the limit as $t \to +\infty$ is 0." Find the error and give the correct result.

SOLUTION

Step 1. Recall the definition for an integral over the whole line: pick a real number a and write

$$\int_{-\infty}^{+\infty} f(x)\,dx = \int_{-\infty}^{a} f(x)\,dx + \int_a^{+\infty} f(x)\,dx,$$

each piece being its own limit, with the two endpoints sent to infinity independently. The integral converges only if both pieces converge.

Step 2. The student computed a single limit $\lim_{t\to\infty}\int_{-t}^{t}$ in which both endpoints move together symmetrically. That is a different quantity (a principal value), not the improper integral. We therefore evaluate the two pieces separately, with $a = 0$.

Step 3. Find an antiderivative. Let $u = x^2+1$, so $du = 2x\,dx$ and $x\,dx = \dfrac{du}{2}$. Then $\displaystyle\int \frac{x}{x^2+1}\,dx = \frac{1}{2}\int \frac{du}{u} = \frac{1}{2}\ln|u| = \frac{1}{2}\ln(x^2+1)$, where the absolute value is dropped because $x^2+1 > 0$.

Step 4. Check: $\dfrac{d}{dx}\left[\dfrac{1}{2}\ln(x^2+1)\right] = \dfrac{1}{2}\cdot\dfrac{2x}{x^2+1} = \dfrac{x}{x^2+1}$.

Step 5. Right piece:

$$\int_0^t \frac{x}{x^2+1}\,dx = \frac{1}{2}\ln(t^2+1) - \frac{1}{2}\ln 1 = \frac{1}{2}\ln(t^2+1).$$

Step 6. As $t \to +\infty$, $t^2+1 \to +\infty$ and so $\ln(t^2+1) \to +\infty$. The right piece diverges to $+\infty$.

Step 7. Since one piece diverges, the whole integral diverges. (The left piece also diverges: $\displaystyle\int_s^0 = 0 - \frac{1}{2}\ln(s^2+1) \to -\infty$ as $s \to -\infty$.)

Step 8. Show why the student's symmetric answer is not meaningful. Send the endpoints to infinity at different rates, for example $-t$ and $2t$:

$$\int_{-t}^{2t} \frac{x}{x^2+1}\,dx = \frac{1}{2}\ln(4t^2+1) - \frac{1}{2}\ln(t^2+1) = \frac{1}{2}\ln\frac{4t^2+1}{t^2+1}.$$

Step 9. Divide numerator and denominator by t^2: $\dfrac{4t^2+1}{t^2+1} = \dfrac{4+\frac{1}{t^2}}{1+\frac{1}{t^2}} \to 4$, so this version tends to $\dfrac{1}{2}\ln 4 = \ln 2 \neq 0$. Different ways of sending the endpoints to infinity give different values, so the symmetric limit 0 says nothing about the improper integral.

Answer. The error is replacing the two independent limits by the single symmetric limit $\int_{-t}^{t}$. The integral actually diverges, because $\displaystyle\int_0^{+\infty} \frac{x}{x^2+1}\,dx = +\infty$